RACHIT JAIN

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Education

Board/Degree	Name of Institution	Year	Score
B-TECH (Computer Software Engg)	Delhi Technological University, Delhi	2026	8.2/10
C.B.S.E. (Class XII)	Kothari International School, Noida	2022	89.4%
C.B.S.E. (Class X)	Kothari International School, Noida	2020	96.6%

Work Experience

SDE Intern, MPS Limited | Noida *[Jan'26–Jun '26]*
- Built the frontend for a Legal Deposit Libraries (LDL) Digital Content Management Platform enabling UK publishers to submit and manage digital publications across six Legal Deposit Libraries, with a public-facing Content Display Platform for library terminal users to discover and read approved content
- Developed a **React**-based Publisher Submissions Portal featuring a multi-step content submission workflow with metadata entry, multi-format file upload (**PDF/ePUB**), and confirmation review pages, a filterable content dashboard with dynamic filter bars, sortable table headers, and paginated results, role-based views for publisher staff and LDL administrators, content editing and reassignment interfaces, and service-layer integration with a **Spring Boot** backend via **RESTful APIs** with **JWT** authentication

SDE Intern, MPS Limited | Noida *[Jan'26–Jun '26]*
- Designed an AI-driven SDLC automation platform that transforms raw requirements documents into complete software development artifacts using specialized **LangGraph** agents with automatic **Pydantic** validation, generating work breakdown structures, user stories with acceptance criteria, technical specifications, non-functional requirements, deployment architectures, sprint plans, test scenarios, and workspace structures
- Implemented an intelligent agent orchestration system with **MCP** host communication via stdin/stdout, featuring **10** specialized AI agents with semantic validation ensuring every user story has corresponding acceptance criteria and all components have defined responsibilities, built with **Java** and **Spring Boot** backend, **React** frontend, **Apache PDFBox** and **Apache POI** for multi-format document parsing, **HuggingFace** for AI model integration, and **GitHub API** for version-controlled artifact storage, generating structured Markdown artifacts with commit tracking

SDE Intern, Gabbit Trans Systems *[Jun'25–Jul '25]*
- Architected a comprehensive health app for the Swasth project, leveraging **machine learning** algorithms and predictive analytics for early detection of disease
- Built and integrated robust features using **Android Studio**, **Flutter**, **Dart**, **Node.js**, **Express.js**, and **JavaScript**, enhancing feature responsiveness by **30%** and improving user engagement metrics by **25%**

AI Intern, Path Infotech Ltd | Noida *[Jun'24–Jul '24]*
- Constructed a high-performance conversational chatbot, fine-tuning the **Llama 3** model with a custom dataset to automate and enhance service tasks for handling AC product defects, achieving a **40%** reduction in manual support
- Implemented advanced **NLP** and machine learning features using **Hugging Face** to improve accuracy in addressing customer queries and reducing average response time by **35%**

Projects

Bharat Bhraman | CLIP+ONNX, Flask, Figma, Supabase, Python, React, TensorFlow, PyTorch, Node.js
- Designed an AI-driven travel platform with real-time itinerary generation using the **Gemini API**, **CLIP**-based image-to-place recommendation, and hybrid similarity fusion (**TF-IDF** + cosine + geo) displaying top **5** most relevant suggestions
- Implemented smart guide matchmaking, hidden gem discovery, group expense splitting and local event detection across **150+** curated locations

Dehaze Rescue Vision | Python, OpenCV, YOLOv8, STTN, DeepSORT
- Designed a real-time de-smoking and de-hazing pipeline using **YOLOv8** (**PyTorch**), Dark Channel Prior (**OpenCV**), and **STTN** inpainting, improving indoor fire scene visibility by **50%** for rescue operations
- Applied **deep learning**, **computer vision**, and tracking (**DeepSORT/ByteTrack**) to reconstruct clear video footage under heavy smoke, enhancing situational awareness and emergency response efficiency

Project & Researchpaper | Wavelet extraction, Spectral kurtosis, ZCR, MEL Spectrogram, SVC, XGBoost, Ensemble Random Forest
- Conducted research on audio-based machine fault diagnosis, utilizing **wavelet extraction**, **spectral kurtosis**, **zero crossing rate**, and **MEL spectrogram** to enhance fault detection accuracy
- Achieved **98.4%** detection accuracy through the application of **SVC**, **XGBoost**, and **ensemble random forest** models with a robust hybrid feature extraction system, improving diagnostic precision and operational efficiency

Projects (contd.)

Automated Brain Tumor Segmentation
- Built an automated glioma segmentation pipeline using a **3D Attention U-Net** with residual blocks, **Squeeze-and-Excitation** attention, and deep supervision on multi-modal MRI (T1, T1ce, T2, FLAIR), achieving a mean Dice of **0.8129** (WT: **0.89**, TC: **0.81**, ET: **0.74**) on BraTS 2018–2020 (**989** volumes)
- Implemented **N4ITK** bias correction, z-score normalization, boundary-focused patch sampling (128°), composite loss (**Focal Tversky** + label-smoothed CE + boundary loss), and **AdamW** with cosine warm restarts for clinically viable tumor delineation

Smart Traffic Signal Optimization | YOLOv8, DeepSORT, OpenCV, Python, NumPy
- Developed a real-time smart traffic signal system that adjusts green signal duration based on vehicle density using object detection (**YOLOv8**) and tracking (**DeepSORT**)
- Simulated four-way traffic with adaptive frame-based vehicle tracking
- Achieved traffic flow efficiency up to **77.16%** over traditional fixed timing

Publications

Audio Based Machine Fault Diagnosis using Hybrid Feature Extraction and Ensemble Learning *(Published)*
International Conference on Computing, Communication and Networking Technologies (15th ICCCNT 2024)
Authors: Rachit Jain, Shashvat Singhal, Shashi Sah

Automated Brain Tumor Segmentation Using Advanced 3D Attention U-Net with Deep Supervision and Boundary-Aware Loss Functions*(Published)*
Journal of Computer Engineering and Applications (JCEA), Volume 16 Issue 5, 2026
Authors: Rachit Jain, Akash Singh

Technical Skills

- **Coursework:** Data Structures and Algorithm, Object Oriented Programming, DBMS, Operating Systems, Computer Networks, Computer Architecture, Compiler Design, Digital Electronics, Software Engineering, Malware Analysis
- **Extra Courses:** Agentic AI, Machine Learning, Deep Learning, Audio Analysis, Natural Language Processing, LLM
- **Programming:** C, C++, Python, Javascript, Java
- **Tools & Frameworks:** Git-GitHub, Android Studio, Google Collab, SQL, PostgreSQL, OpenCV, Node.js, Express.js, React, LangChain, LangGraph

Additional

- **Head, AIMS Society (DTU)** – Led teams to design and execute **AI/ML-based projects and competitions**, mentored members in implementing **real-world algorithms**, and expanded **industry partnerships** to enable exclusive **internships and research opportunities**.
- **Languages:** English, French, Hindi